# Hydrogen Energy Levels: Quantised Transitions

Hydrogen Energy Levels · oPhysics

| Senior Secondary (Years 11-12) | ~30 minutes | Science · Physics |
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Simulation: https://ophysics.com/m1.html
Curriculum links: quantum physics

## Learning intentions

- I can describe the energy levels of the hydrogen atom as quantised (discrete) rather than continuous.
- I can relate an electron transition between levels to the energy of an emitted or absorbed photon.
- I can calculate a photon's wavelength from its energy.

## Getting started

Open the simulation and examine the energy level diagram for hydrogen, noting the energy value of levels n = 1, 2 and 3.

## Tasks

**1.** Record the energy values of the n = 1, n = 2 and n = 3 levels shown in the diagram.

**2.** Calculate the energy released when an electron transitions from n = 3 down to n = 1.

*Working:*

**3.** Using $E = hc/\lambda$, calculate the wavelength of the photon released in Task 2 (h = 6.63 × 10■³■ J·s, c = 3.00 × 10■ m/s). Which part of the electromagnetic spectrum does this fall into?

*Working:*

**4.** Of all the transitions down to n = 1 shown in the diagram, which one releases the highest-energy photon? Explain how you can tell from the diagram alone, without calculating.

## Extension (optional)

Explain why the hydrogen atom's emission spectrum consists of a small number of sharp lines, rather than a continuous rainbow of colours.